

阶段 1: 拥塞检测 (epoch τ)

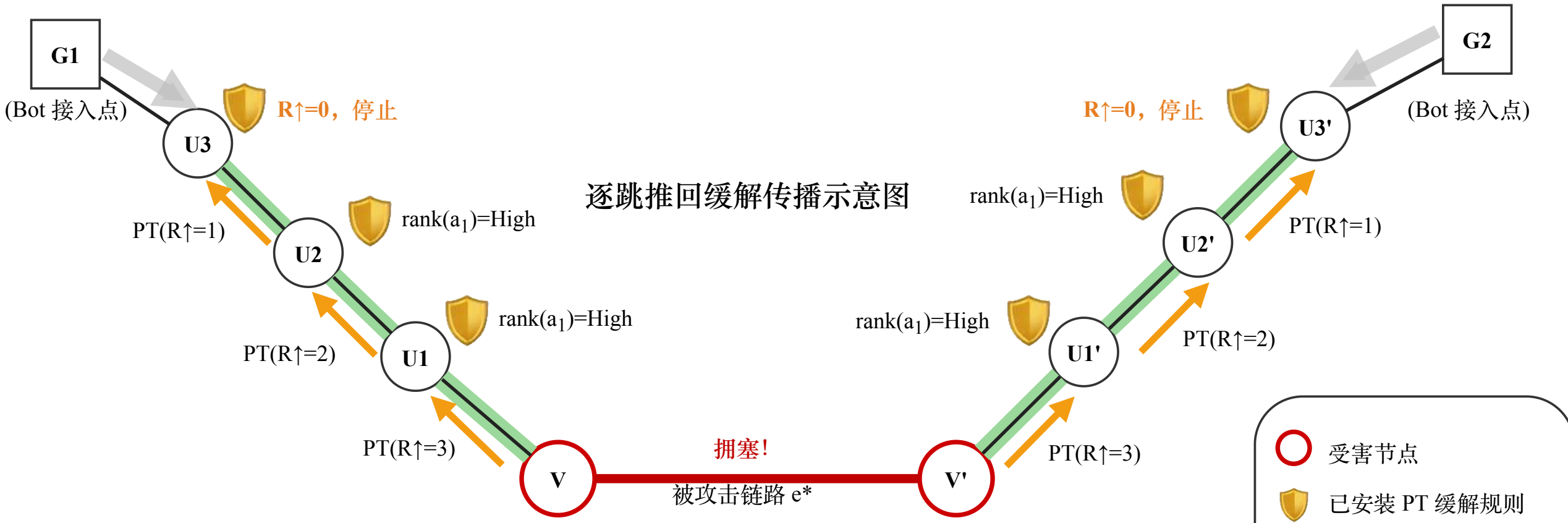
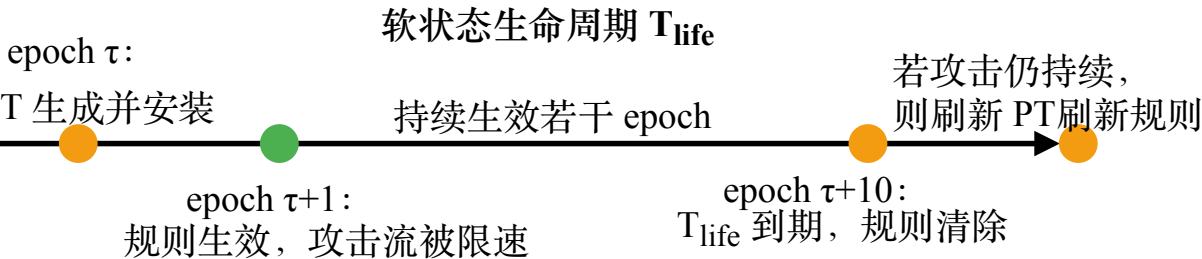
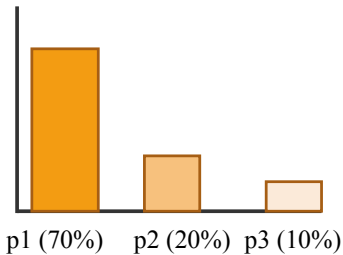
节点 V 数据面检测: $q_{v,e^*} > Q_{th}$, 故 $c(v,e^*)=1$

Top-K 候选 key: a_1, a_2

入端口贡献分布 $\pi(a_1,p)$:

$p_1=70\%$, $p_2=20\%$, $p_3=10\%$

$S^{net}(a_1)=0.87 \geq \Theta_{pb}=0.7$, 触发推回



阶段 3: 缓解效果

推回前: Bot → G → U₃ → U₂ → U₁ → V → e*, h=5 跳, $W_{waste} = rate \times 5$

推回后: Bot → G → U₃ 邻处被限速/丢弃, h=2 跳, $W_{waste} = rate \times 2$

W_{waste} 从 5 降到 2, 降低 60%

中间三跳链路带宽释放给良性流量